Lesson plan · for the teacher

Data in class: record, chart, decide

IoT · Level 1 · Lesson 4

Time
40 min

Materials
Board and chalk, with a ruler for straight bars; A week of rain readings: your class gauge log, or a sample week; Squared paper or exercise books, for pairs; Worksheet: Rain data detective (one per pair, optional)

Learning goals

- Record a week of rain readings in a table with units.
- Draw a bar chart with equal steps and labelled axes.
- Spot a pattern and use a class rule to make a decision.

Activity: Our class rainfall chart: Chart

1. Warm-up (5 min): Ask: “When should we water the school garden?” Collect answers — every day? When it looks dry? Write “How could data help us decide?” on the board.
2. Record (7 min): Pupils copy the table — Day, Time, Rain (mm) — and fill it from the class gauge log or the sample data. Discuss what to write for a day with no reading.
3. Chart (13 min): Draw the axes together: days along the bottom, “Rain (mm)” up the side in equal steps of 2. Pairs draw their bars on paper while volunteers draw them on the board.
4. Decide (10 min): Ask pairs to name one pattern. Then agree a class rule, such as “water if less than 5 mm fell in three days and the soil feels dry”. Apply it to the chart.
5. Wrap-up (5 min): Explain that connected devices can do these steps automatically with sensors and rules. Exit ticket: the four steps in order, and today’s decision.

Draw the empty table and chart axes on the board. Sample data if you have no gauge yet (8 am readings, made up for practice): Mon 14 mm, Tue 9, Wed 0, Thu 0, Fri 0, Sat 0, Sun 2.

Discussion

- Why not leave it blank?
- What does the chart tell us?
- Does our rule say water today, or wait?

Exit ticket

- The four data steps in order, and our decision today.

Try it in class tomorrow
Chart a week of rain: Copy a week of rain readings onto the board — your class gauge or the sample in the lesson plan. Pupils draw a bar for each day, name one pattern, and vote on a watering rule. Check the rule against the chart together.

Worksheet · for pupils

Data in class: record, chart, decide

IoT · Level 1 · Lesson 4

Name _____ Class _____ Date _____

1 **Put the steps for using your class rain data in order.**

- [ ] Spot the pattern in the bars
- [ ] Decide, using the class rule
- [ ] Record each reading in a table
- [ ] Draw a bar chart, one bar per day

2 **A week of 8 am rain readings (mm): Mon 14, Tue 9, Wed 0, Thu 0, Fri 0. Which pattern matters most for watering the garden?**

- ◯ Three dry days in a row since Tuesday
- ◯ The week's total was 23 mm
- ◯ Monday was the wettest day

3 **A blank box in a rain table means it didn't rain.**

◯ True ◯ False

4 **Which of these make a class bar chart easy to read?**

- ☐ Bars of different widths, for variety
- ☐ The day written under each bar
- ☐ Leaving out the dry days
- ☐ A label on the side: "Rain (mm)"
- ☐ Equal steps up the side: 0, 5, 10, 15

5 **Match each data step to what a connected garden system does.**

Record	• • • •	A graph appears on a phone screen
Chart	• • • •	Readings are saved in a table automatically
Spot the pattern	• • • •	A rule switches the pump on
Decide	• • • •	Software flags several dry readings in a row

Answers · for the teacher

Data in class: record, chart, decide

IoT · Level 1 · Lesson 4

1 **Top to bottom: 3, 4, 1, 2**

Record, chart, spot, decide. Each step needs the one before — and connected devices follow the same order automatically.

2 **Three dry days in a row since Tuesday**

A run of dry days is the pattern that tells you the soil may need water.

3 **False**

A blank could mean no rain or no reading.

4 **A label on the side: "Rain (mm)"; Equal steps up the side: 0, 5, 10, 15; The day written under each bar**

The side label says what the bar heights mean, equal steps stop bars looking taller or shorter than they should, and the day under each bar shows when the rain fell.

5 **Record — Readings are saved in a table automatically**
Chart — A graph appears on a phone screen
Spot the pattern — Software flags several dry readings in a row
Decide — A rule switches the pump on

Same four steps, done by machines. Knowing them helps you and your pupils check whether an automatic decision makes sense.

✎ **For you to think about**

What decision in your school could your pupils make with data they collect themselves? What would they record?